

Proactive Physical Safety Reasoning for Robot Manipulation

Abstract—Physical safety is a critical concern for embodied AI systems operating in real-world environments, especially in factory settings where human-robot collaboration introduces high-stakes interaction risks. Robots must reason about spatial configurations, dynamic motion, and material properties to proactively prevent accidents. We introduce a novel benchmark framework for proactive physical safety reasoning that integrates vision-language models (VLMs) with structured risk classification and visual grounding. Our approach focuses on three core types of physical risk—static spatial conflicts, dynamic motion hazards, and functional tool misuse—captured through a multi-stage reasoning pipeline. The system generates context-aware prompts for precise object grounding using Grounding DINO and SAM2, enabling identification of safety-critical entities and anticipation of unsafe outcomes. Importantly, the framework is designed to mitigate foundation model inference latency through early risk detection and predictive reasoning, allowing for timely safety interventions in fast-moving collaborative environments. To support robust evaluation, we propose a comprehensive data collection strategy using industrial footage, accident reports, and simulated interactions, spanning household, outdoor, and especially factory scenarios. This work lays essential groundwork for trustworthy and safe embodied AI by empowering robots to understand, anticipate, and act to prevent physical risks through commonsense visual reasoning.

I. INTRODUCTION

The deployment of robots in unstructured human environments, homes, factories, and outdoor settings, introduces unique safety challenges that distinguish embodied AI from purely digital systems. When a robot manipulates objects in the physical world, its decisions have immediate and potentially irreversible consequences. A misidentified object, a poorly timed movement, or a failure to recognize spatial conflicts can result in property damage, human injury, or task failure.

Current robotic safety approaches are primarily *reactive*: they detect hazards only when they are already present in sensor data. This reactive paradigm is insufficient for real-world deployment, where safety margins are thin and correction time is limited. Consider a robot arm approaching a human worker: by the time contact is detected, injury may be unavoidable. What is needed is *proactive reasoning*: the ability to anticipate risks before they materialize and to adjust behavior accordingly.

Recent advances in vision-language models (VLMs) offer a promising foundation for such capabilities. Models like GPT, Gemini, and open-source alternatives demonstrate remarkable commonsense understanding of visual scenes, including spatial relationships, object properties, and human behavior. However, applying these models to safety-critical robotics requires addressing fundamental challenges:

- **Inference latency:** VLMs can take hundreds of milliseconds to several seconds to process visual inputs,

which is too slow for real-time safety decisions.

- **Lack of structured risk reasoning:** VLMs excel at open-ended description but lack frameworks for systematic physical safety assessment.
- **Grounding precision:** Generic object detection may miss safety-critical details like protective equipment, object stability, or human attention states.
- **Benchmark scarcity:** No existing datasets systematically evaluate physical safety reasoning across diverse real-world scenarios.

This paper addresses these challenges through four main contributions:

- **A structured risk taxonomy** for physical safety in robot manipulation, categorizing hazards into static spatial conflicts, dynamic motion risks, and functional tool misuse.
- **A multi-stage reasoning pipeline** that leverages VLMs for context-aware scene analysis, generates targeted grounding prompts, performs risk prediction, and suggests preventive actions, all while mitigating inference latency through early classification and predictive buffering.
- **An integration framework** with vision foundation models (Grounding DINO, SAM2) that enables precise localization of safety-relevant entities based on semantic understanding rather than generic detection.
- **A benchmark design and data collection strategy** drawing from accident reports, industrial footage, and simulation to create a diverse evaluation set spanning daily life, household, outdoor, and factory scenarios.

Our approach is inspired by the three-layer safety framework proposed for robotic systems (high-level goal safety, physical safety, and low-level actuator safety). We focus specifically on the middle layer, *physical safety*, which is unique to embodied AI and directly addresses the real-world consequences of robotic actions. By combining commonsense reasoning from VLMs with structured safety assessment, we enable robots to reason about risks in ways that go beyond rule-based systems while maintaining the rigor needed for safety-critical applications.

II. RELATED WORK

A. Robot Safety and Risk Assessment

B. Vision-Language Models for Robotics

Existing applications primarily focus on task execution rather than safety assessment. Recent work on vision-language models for safety has examined adversarial inputs and model robustness, but not physical safety reasoning in embodied contexts.

TABLE I
PHYSICAL RISK TAXONOMY FOR ROBOT MANIPULATION

Risk Type	Description & Examples
Static	Conflicting spatial relationships in the absence of motion.
Dynamic	Risks arising from motion and timing.
Functional	Improper tool use or mechanical interaction.

C. Benchmarks for Robotic Safety

Existing robotics benchmarks focus primarily on task completion such as manipulation skill, navigation. Safety is typically evaluated indirectly through task success rates or constraint violations.

III. PROBLEM FORMULATION

A. Physical Safety Layer

We adopt a three-layer framework for robotic safety, focusing specifically on the *physical safety layer*: (1) High-level safety addresses goal alignment and ethical considerations; (2) Physical safety concerns spatial configurations, dynamics, and material interactions in the environment; (3) Low-level safety manages actuator limits and motor health. Our work targets Layer 2, which is unique to embodied systems and directly impacts real-world outcomes.

B. Risk Taxonomy

We define three core categories of physical risk, derived from analysis of accident reports and safety guidelines. Table I summarizes these categories.

These categories are not mutually exclusive, many scenarios exhibit multiple risk types simultaneously (e.g., a person without gloves reaching toward a moving drill exhibits both dynamic and functional risks).

C. Task Definition

Given a visual input I (image or video sequence) and optional context C (text description, robot intent), the system must:

- 1) Identify safety-relevant entities and their spatial relationships
- 2) Classify risk types present in the scene
- 3) Predict potential unsafe outcomes in the near future (1-3 seconds)
- 4) Generate natural language explanation of identified risks
- 5) Suggest preventive actions (spatial adjustment, temporal control, communication)

Critically, the system must operate under real-world constraints including VLM inference latency (hundreds of milliseconds to seconds) and the need for timely intervention before risks materialize.

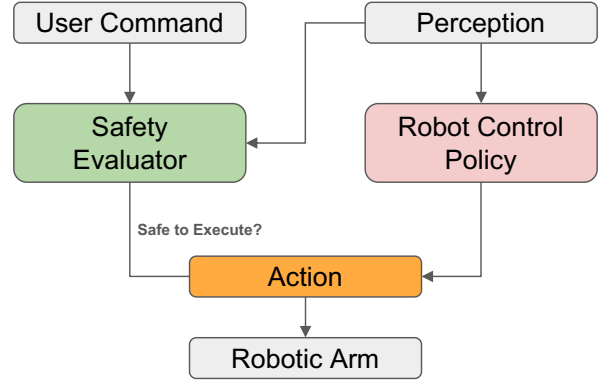


Fig. 1. Proactive Safety Reasoning Pipeline

IV. METHOD: PROACTIVE SAFETY REASONING PIPELINE

Our framework combines vision-language reasoning with structured safety assessment through a five-stage pipeline. Each stage is designed to mitigate VLM latency while enabling rich commonsense understanding of physical risks. Figure 1 illustrates the overall architecture.

A. Stage 1: Context-Aware Scene Analysis and Grounding Keyword Generation

Rather than using fixed object detection queries, we employ a VLM to analyze the scene and reason about which entities are safety-relevant. This stage performs two critical functions:

- 1) **Scene understanding:** The VLM identifies humans, tools, robot components, and environmental features relevant to physical safety.
- 2) **Grounding keyword generation:** Based on this analysis, the VLM produces precise phrases optimized for Grounding DINO (e.g., “unprotected human hand”, “sharp blade”, “moving robot arm”).

The system prompt for this stage emphasizes commonsense reasoning about human states. For example, we instruct the model: “*You are a safety-aware visual reasoning agent. Analyze the provided image and identify which objects, humans, or tools are safety-relevant... Use commonsense and context. For example: a person in casual clothes is more vulnerable than one with full protective gear; a tilted pot near the edge may cause a spill.*”

Output includes a scene summary, per-object descriptions with keywords, and a combined grounding query. This approach provides three key advantages over generic object detection:

- **Semantic filtering:** Only safety-relevant entities are selected for grounding
- **Contextual specificity:** Keywords adapt to scene details (e.g., “gloved hand” vs “bare hand”)
- **Latency mitigation:** This lightweight reasoning step (typically <1 second) enables immediate start of parallel grounding

B. Stage 2: Visual Grounding and Segmentation

The generated grounding keywords are fed to Grounding DINO for object detection, producing bounding boxes with confidence scores. These boxes serve as prompts for SAM2, which generates precise segmentation masks. For video inputs, SAM2’s temporal propagation tracks objects across frames.

This stage operates in parallel with subsequent reasoning stages, further reducing end-to-end latency. While segmentation proceeds, the VLM can begin risk classification using the scene analysis from Stage 1.

C. Stage 3: Risk Classification and Outcome Prediction

Given the scene analysis and (optionally) preliminary grounding results, the VLM classifies risk types and predicts potential outcomes. The system prompt incorporates our risk taxonomy and emphasizes temporal anticipation:

“Consider three core risk types: *Static (conflicting spatial relationships)*, *Dynamic (movement risk)*, *Functional (improper tool use)*. Use commonsense reasoning: *trained worker with safety gear* → lower risk; *distracted person* → higher risk. Predict short-term outcomes (1-3 seconds) using causal inference.”

Output includes risk type labels, confidence scores, affected objects, textual descriptions, and anticipated outcomes. This stage is *proactive*, it predicts what *could* happen rather than just describing what *is* visible. Early risk classification enables immediate conservative fallback policies (e.g., pause robot motion) even before full VLM output is available.

D. Stage 4: Natural Language Explanation

The system generates human-understandable explanations that reference specific objects, spatial relationships, and contextual factors. This serves multiple purposes: (1) transparency for human operators, (2) debugging and error analysis, (3) training data for downstream models. The explanation includes both risk rationale and preventive suggestions grounded in physical commonsense.

E. Stage 5: Policy Adjustment Recommendation

The final stage translates risk assessment into actionable robot behavior modifications:

- **Spatial adjustments:** Modify trajectory to avoid collision zones
- **Temporal adjustments:** Pause, slow down, or delay motion
- **Communication:** Alert human operators to high-risk situations

Policy recommendations balance safety and task goals. For example, pausing is preferred when risk is high but uncertainty exists; trajectory adjustment is used when safe alternatives are available.

F. Platform Goal and Evaluation Loop

This project builds the *execution and evaluation platform* required to study proactive physical safety in human-robot collaborative manipulation. In our setting, a behavior policy π_{PPO} proposes robot actions to accomplish a manipulation task, while a separate safety evaluation module (developed in parallel) determines whether executing the proposed action is safe when a human is present. Concretely, at control step t the platform runs:

- 1) state acquisition and task observation o_t ,
- 2) behavior action proposal $a_t = \pi_{\text{PPO}}(o_t)$,
- 3) safety decision $g_t \in \{0, 1\}$ from the safety evaluation module,
- 4) execution: if $g_t = 1$ execute a_t ; otherwise **stop** the robot for that step.

The platform is designed to operate identically in simulation and real environments, enabling controlled scenario replay and consistent logging for safety evaluation.

G. Simulation Environment in Isaac Lab

We implemented the complete manipulation environment in NVIDIA Isaac Lab, with a UR10e robot arm and parallelizable simulation rollout support. The simulation stack includes: (i) robot and gripper model integration, (ii) a Cartesian action interface with inverse-kinematics (IK) control, (iii) task assets and reset/randomization utilities. Figure 2 shows the simulation platform in Isaac Lab.

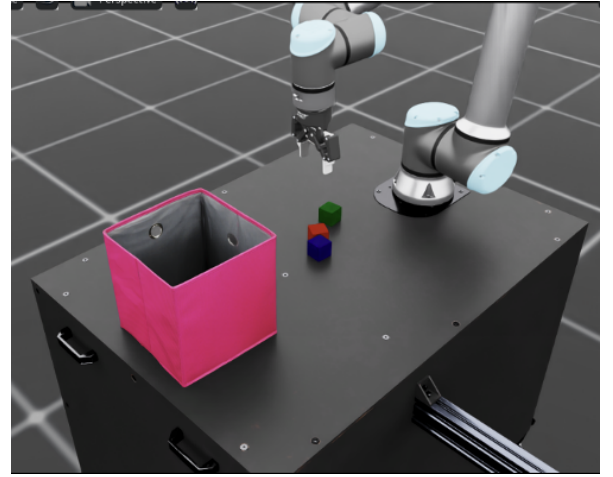


Fig. 2. Simulation Environment in Isaac Lab

Robot and Control Interface. The behavior policy outputs a *Cartesian target pose command* for the gripper:

$$a_t = [x_t^g, y_t^g, z_t^g, \psi_t^g], \quad (1)$$

where (x_t^g, y_t^g, z_t^g) is the desired gripper position and ψ_t^g is the desired yaw. At runtime, an IK controller converts the target into joint-level commands for the UR10e. The control frequency is fixed at **20 Hz**, matching the intended real-world execution rate.

Task Assets and Setup. The task is a pick-and-place scenario with **three cubes** (red, green, blue) placed randomly

on a tabletop and a container as the placement target. The platform supports either (i) sequential completion (one cube at a time) or (ii) full-task completion within one episode (place all cubes). This flexibility is useful for isolating safety-critical phases (approach, grasp, transit, place) during scenario design. Figure 3 demonstrated the 4 safety-critical phases.

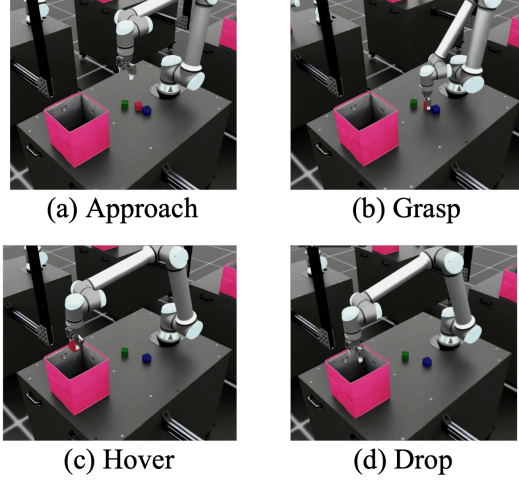


Fig. 3. Safety-critical phases (approach, grasp, transit, place)

Observation Space. In the current milestone, we use a compact state representation to prioritize platform readiness and safety-gating integration. The observation for the target object is:

$$o_t = [x_t^{\text{cube}}, y_t^{\text{cube}}, z_t^{\text{cube}}, \psi_t^{\text{cube}}], \quad (2)$$

containing the cube 3D position and yaw. (Vision-based observations are intentionally excluded in this platform milestone to reduce confounds during safety evaluation development.)

H. Behavior Policy Training (PPO) and Reward Design

The behavior policy π_{PPO} is trained to complete pick-and-place via reinforcement learning. The platform provides reward computation, termination handling, parallel simulation rollouts, and hooks for policy evaluation under safety gating.

Reward Function. We implement a shaped reward with four components encouraging (i) approaching the cube, (ii) lifting, (iii) hovering over the container, and (iv) successful placement:

$$r_{\text{approach}} = 1 - \tanh(10 \cdot d_{\text{eff} \rightarrow \text{cube}}), \quad (3)$$

$$r_{\text{lift}} = \tanh(10 \cdot h_{\text{cube}}), \quad (4)$$

$$r_{\text{hover}} = 1 - 10 \cdot \tanh(d_{\text{cube} \rightarrow \text{container}}^{xy}), \quad (5)$$

$$r_{\text{success}} = \mathbb{I}[\text{cube in container}], \quad (6)$$

where $d_{\text{eff} \rightarrow \text{cube}}$ is end-effector distance to the cube, h_{cube} is cube height above the table, and $d_{\text{cube} \rightarrow \text{container}}^{xy}$ is XY-plane distance from the cube to the container target region. The total reward is the sum of these terms (with optional weights configurable for training stability).

Execution Semantics with Safety Gating. The safety module may veto execution of a_t at any timestep. In the current platform design, a veto triggers a **stop** intervention (i.e., hold/zero-motion command at that step). This creates a clean separation between behavior optimization (PPO) and safety evaluation, and enables interpretable platform-level metrics such as intervention rate, intervention duration, and task success degradation due to safety constraints.

I. Real-World Environment Setup

A corresponding real-world environment has been assembled to mirror the simulation scene and action interface. The setup includes the UR10e manipulator, a parallel gripper matching the simulation end-effector, a tabletop workspace, a container target, and the three colored cubes. The real system is intended to follow the same high-level loop as simulation:

- behavior action in Cartesian target pose space,
- IK-based joint control for execution,
- safety-gated execution at 20 Hz,
- unified logging for post-hoc safety analysis.

This parity is crucial for validating whether safety evaluation policies and benchmarks developed in simulation remain meaningful when transferred to real environments. Figure 4 shows the simulation platform in Isaac Lab.

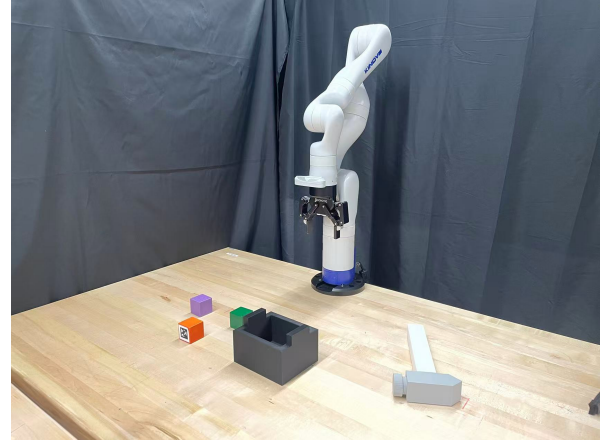


Fig. 4. Real-World Environment Setup

J. Next Milestone: Human Motion Generation in Simulation

The primary remaining milestone for the platform is to generate **safety-critical human motion** in simulation. This component is essential to stress-test the safety evaluation policy under realistic and adversarial conditions. The near-term plan is to implement a scenario library that can systematically vary:

- **timing** (human enters during approach vs. during transit vs. during placement),
- **proximity** (near-miss vs. direct interference),
- **speed** (slow reach vs. sudden intrusion),
- **trajectory archetypes** (cross-path, reach-in, linger near container).

The scenario library will support repeatable evaluation and controlled distribution shifts (train vs. test) to assess generalization of safety policies.

V. CONCLUSION

REFERENCES